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EEE 465/591: Photovoltaics Energy Conversion
Practice Problem Set #1 - Solutions

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1. Unit Classification

Identify if each of the following are a unit of energy, power or other:

> **a. Watts** → *Power* (P = W)
>
> **b. Joules** → *Energy* (E = J)
>
> **c. Watt-hours** → *Energy* (E = W×h = J)
>
> **d. Quad** → *Energy* (1 Quad = 10^{15} BTU)
>
> **e. BTU** → *Energy* (British Thermal Unit)
>
> **f. Horsepower** → *Power* (1 HP ≈ 746 W)

2. Energy-Wavelength Conversions

Convert between energy and wavelength using the photon energy equation:

$$\mathbf{E = hc/\lambda}$$

where $h = 6.626 \times 10^{-34}$ J·s, $c = 3 \times 10^{8}$ m/s

$$\mathbf{E(eV) = 1240/\lambda(nm)} \text{ (convenient form)}$$

> **a. 1 eV → wavelength**
>
> $\lambda = 1240/E = 1240/1 =$ **1240 nm**
>
> **b. 405 nm → energy**

> $E = 1240/\lambda = 1240/405 =$ **3.06 eV**
>
> **c. 0.5 micron = 500 nm → energy**
>
> $E = 1240/500 =$ **2.48 eV**
>
> **d. 2.2 eV → wavelength**
>
> $\lambda = 1240/2.2 =$ **564 nm**
>
> **e. 600 nm → energy**
>
> $E = 1240/600 =$ **2.07 eV**
>
> **f. Si bandgap at 300K ≈ 1.12 eV → wavelength**
>
> $\lambda = 1240/1.12 =$ **1107 nm**
>
> **g. GaAs bandgap at 300K ≈ 1.42 eV → wavelength**
>
> $\lambda = 1240/1.42 =$ **873 nm**

3. Monochromatic Light Source Parameters

Fill in the missing parameters using the relationship between photon energy, flux, and power density:

$$\textbf{Power Density} = \textbf{Photon Energy} \times \textbf{Photon Flux}$$
$$\mathbf{P = E \times \Phi}$$

Photon Energy	Photon Flux	Power Density
1.43 eV	1×10^{19} photons $m^{-2}s^{-1}$	**229 W/m²**
1.12 eV	**4.46×10^{21} photons $m^{-2}s^{-1}$**	800 W/m²
1.25 eV	5×10^{21} photons $m^{-2}s^{-1}$	1000 W/m²

> **Row 1 calculation:**
> $P = E \times \Phi = 1.43$ eV $\times 1 \times 10^{19}$ photons/m²s
> $P = 1.43 \times 1.602 \times 10^{-19}$ J $\times 1 \times 10^{19}$ /m²s = **229 W/m²**

Row 2 calculation:

$\Phi = P/E = 800\ \text{W/m}^2 \div (1.12 \times 1.602 \times 10^{-19}\ \text{J})$

$\Phi = 800 \div (1.794 \times 10^{-19}) =$ $\mathbf{4.46 \times 10^{21}}$ **photons** $\mathbf{m^{-2}s^{-1}}$

Row 3 calculation:

$E = P/\Phi = 1000\ \text{W/m}^2 \div (5 \times 10^{21}\ \text{photons/m}^2\text{s})$

$E = 2 \times 10^{-19}\ \text{J} = 2 \times 10^{-19} \div 1.602 \times 10^{-19} =$ **1.25 eV**